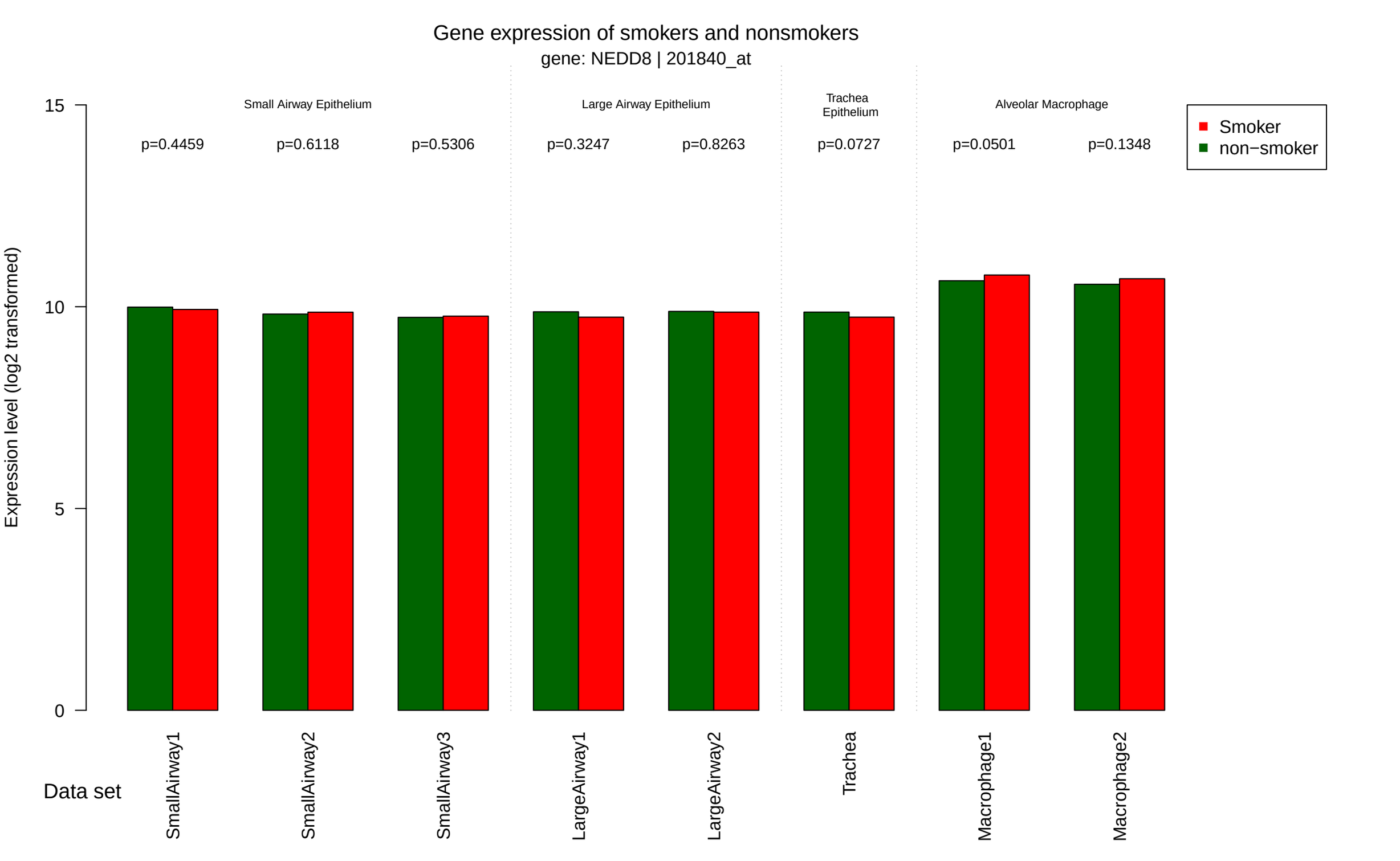
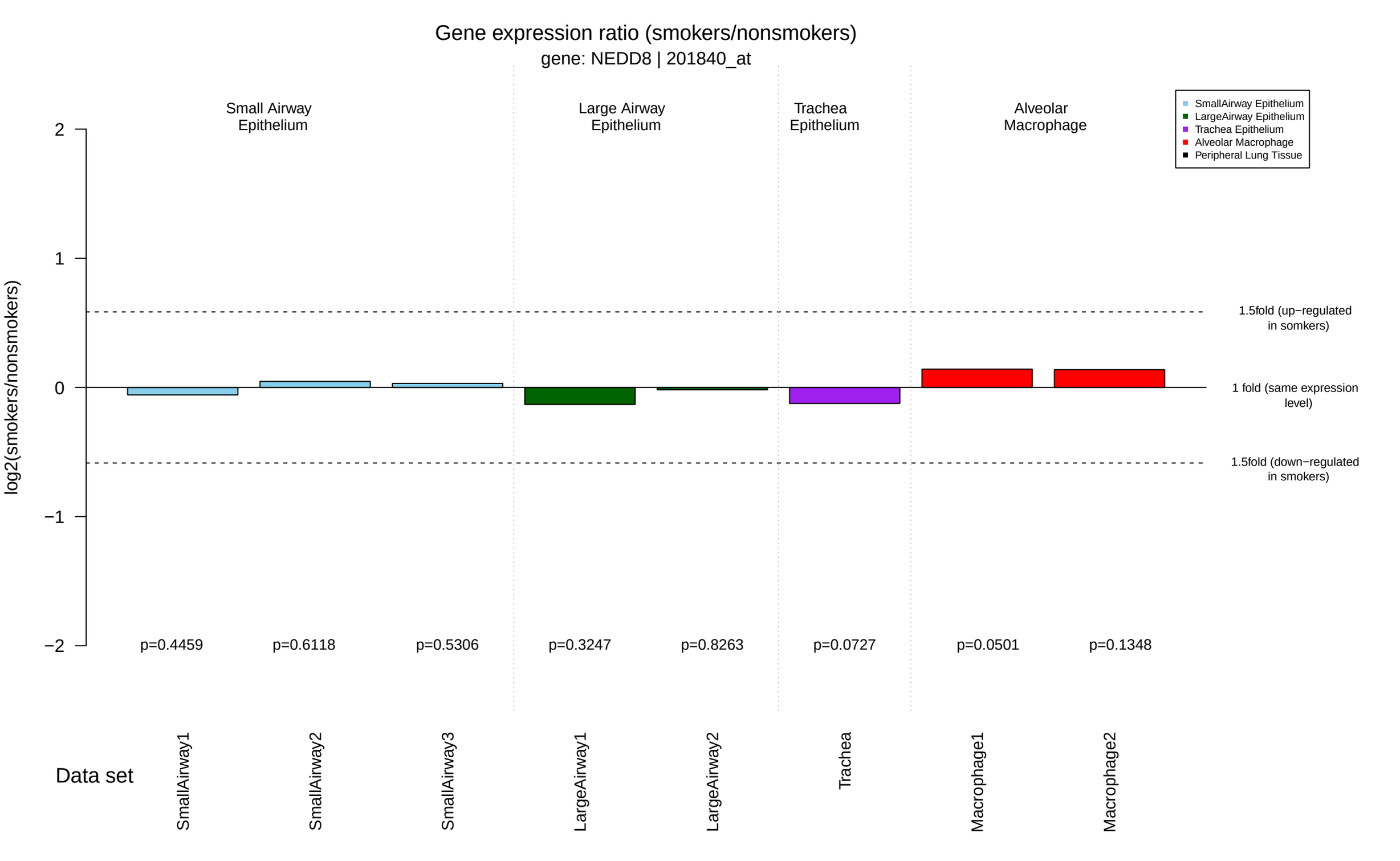
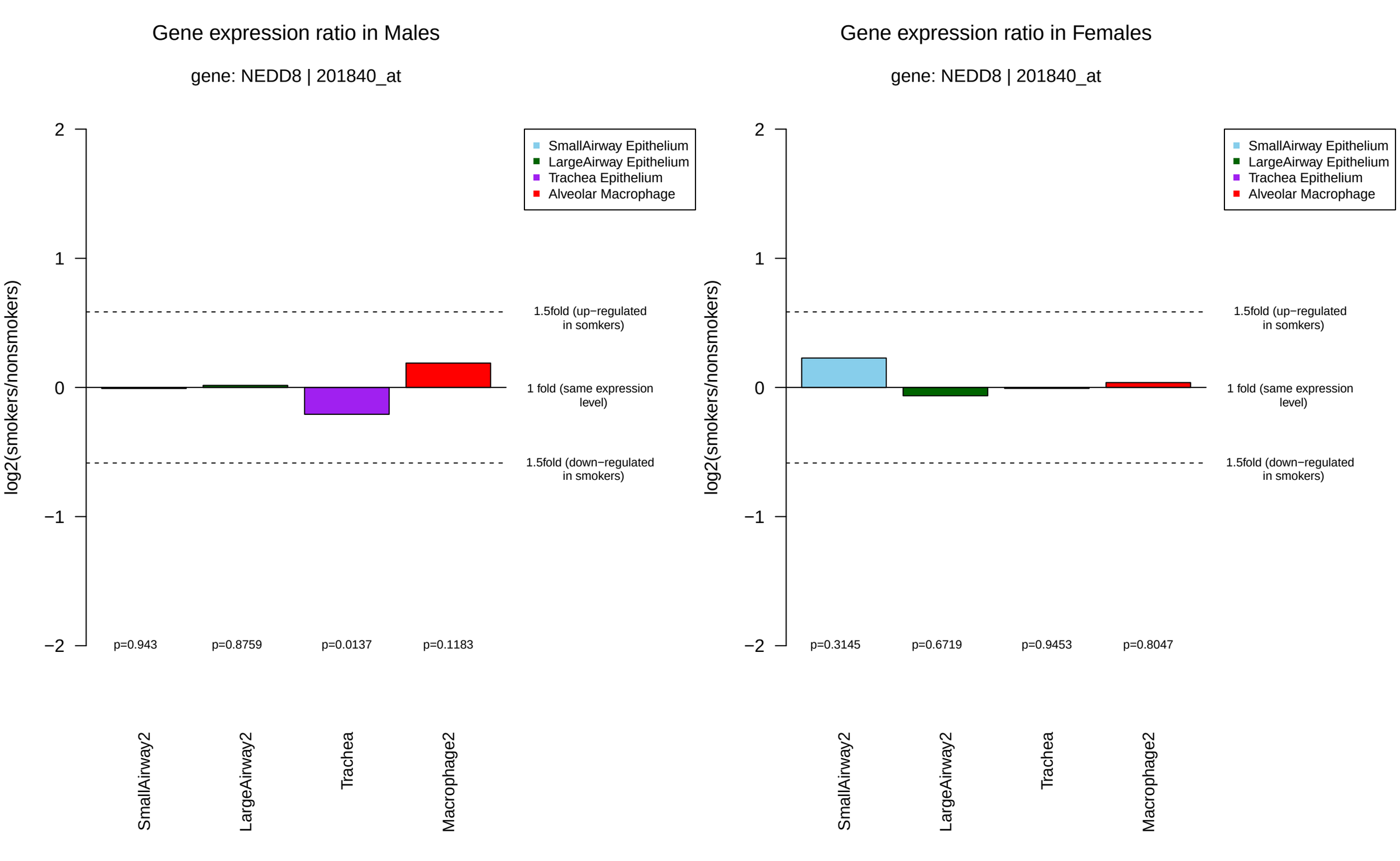
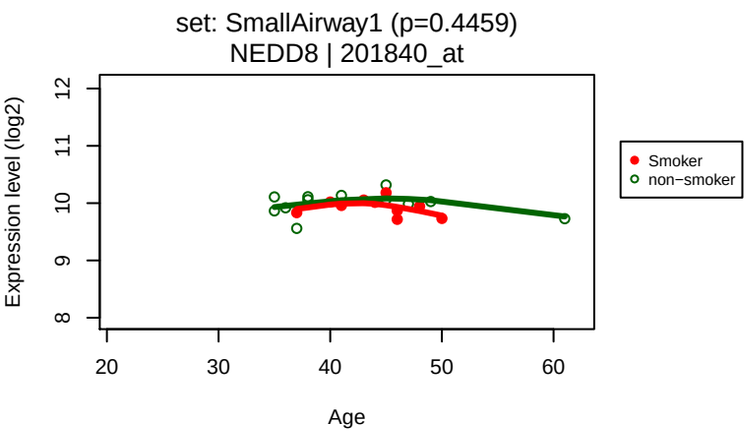
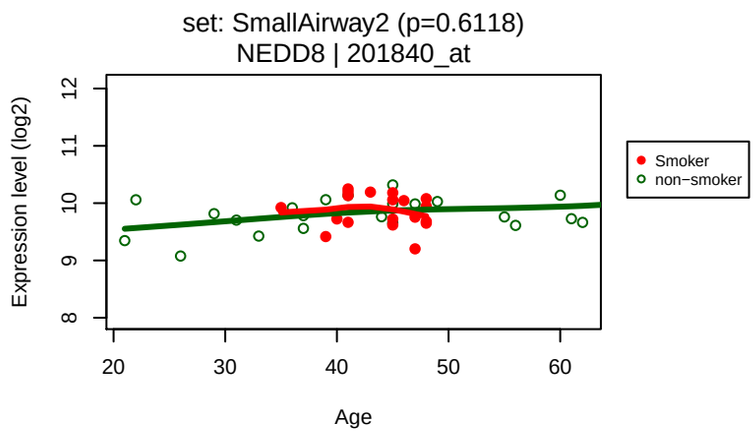
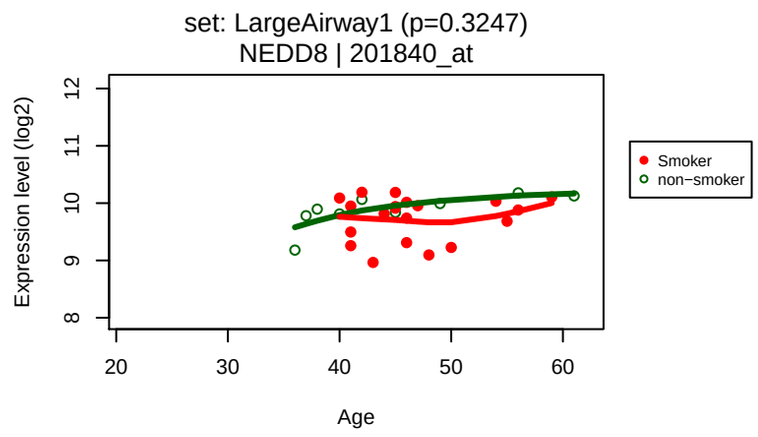

Gene expression of smokers and nonsmokers
gene: NEDD8 | 201840_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4459
p=0.6118
p=0.5306
p=0.3247
p=0.8263
p=0.0727
p=0.0501
p=0.1348
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

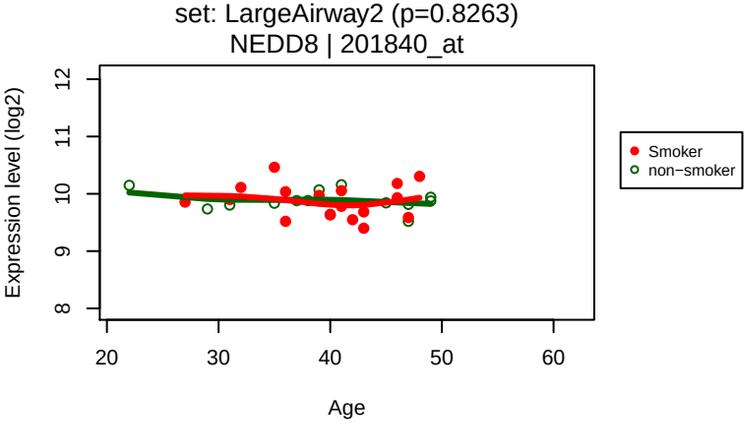

Gene expression ratio (smokers/nonsmokers)
gene: NEDD8 | 201840_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4459
p=0.6118
p=0.5306
p=0.3247
p=0.8263
p=0.0727
p=0.0501
p=0.1348
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

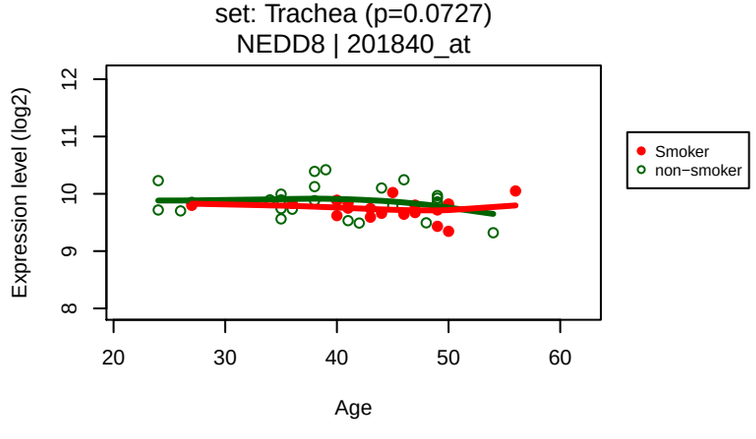

Gene expression ratio in Males
gene: NEDD8 | 201840_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.943
p=0.8759
p=0.0137
p=0.1183
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NEDD8 | 201840_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3145
p=0.6719
p=0.9453
p=0.8047
SmallAirway2
LargeAirway2
Trachea
Macrophage2

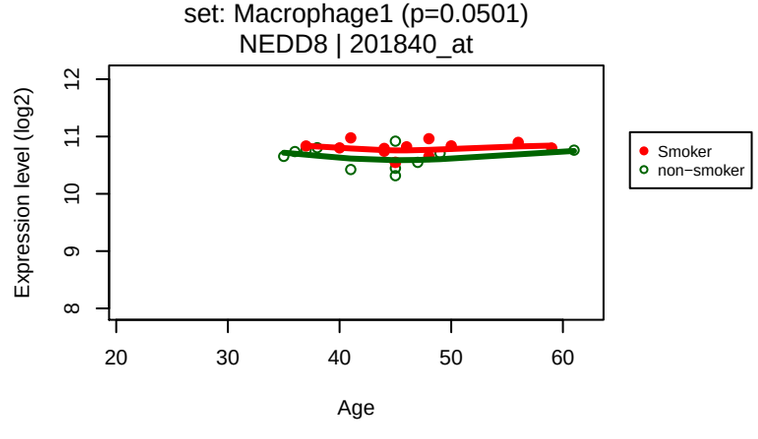

set: SmallAirway1 (p=0.4459)
NEDD8 | 201840_at
Expression level (log2)
Age
Smoker
non−smoker

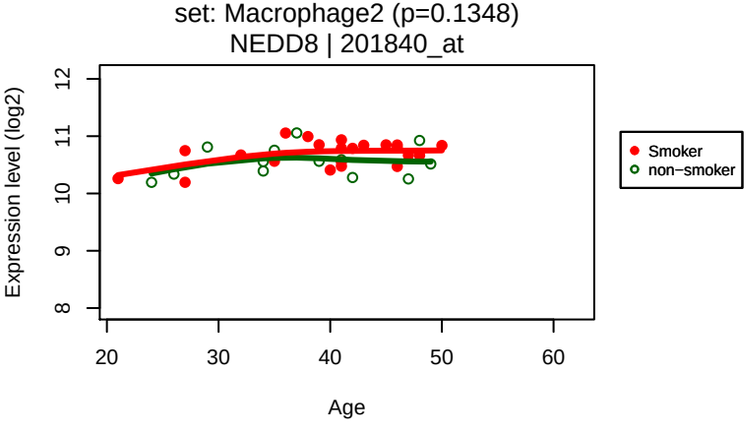

set: SmallAirway2 (p=0.6118)
NEDD8 | 201840_at
Expression level (log2)
Age
Smoker
non−smoker

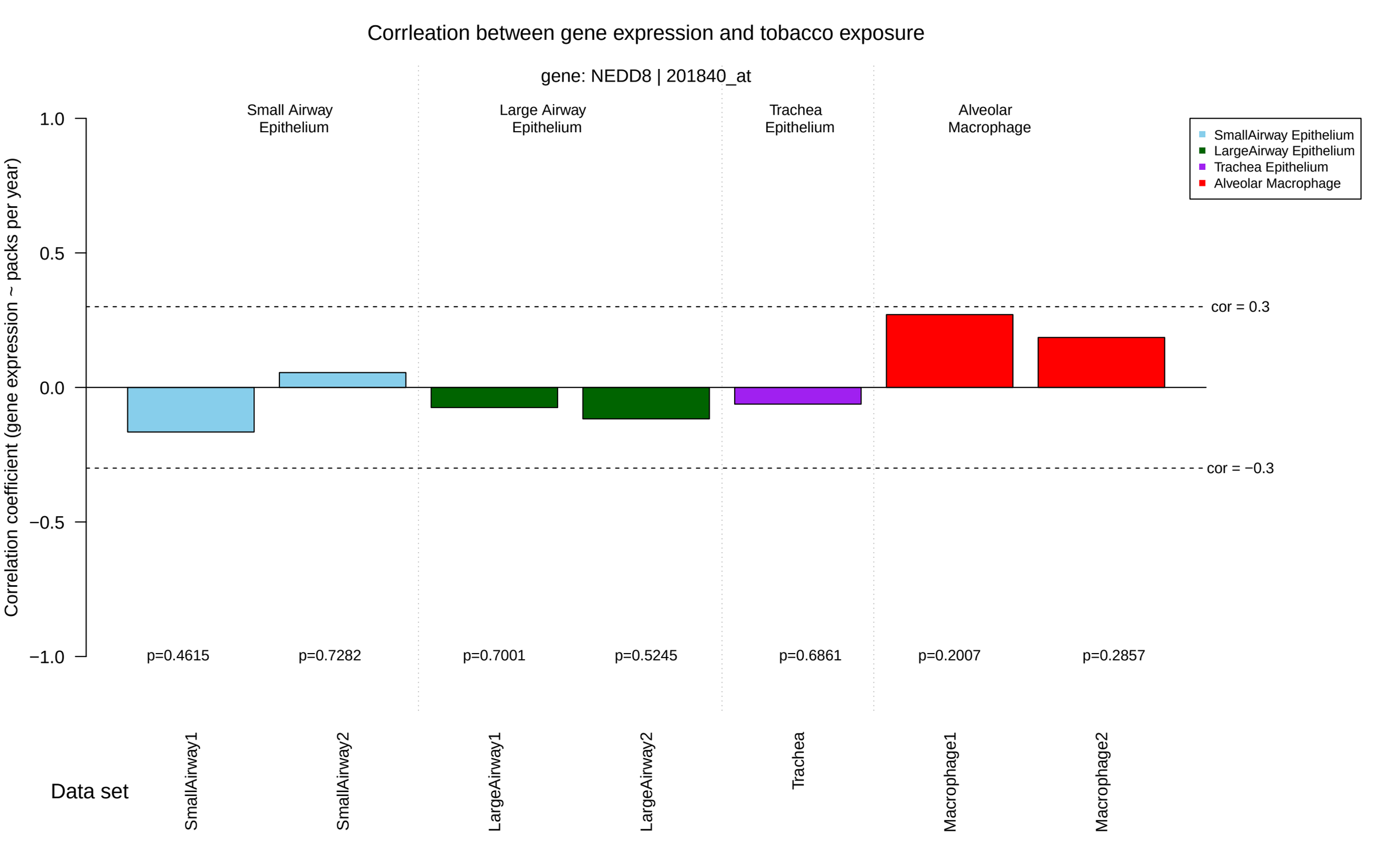

set: LargeAirway1 (p=0.3247)
NEDD8 | 201840_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8263)
NEDD8 | 201840_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0727)
NEDD8 | 201840_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0501)
NEDD8 | 201840_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1348)
NEDD8 | 201840_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NEDD8 | 201840_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4615
p=0.7282
p=0.7001
p=0.5245
p=0.6861
p=0.2007
p=0.2857
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2